

Economic assessment and grid impact of different sharing keys in collective self-consumption

Kjersti Berg^{*†}, Rubi Rana[†], Henning Taxt[†], Marthe F. Dyrne^{*}

^{*}Department of Electric Energy, Norwegian University of Science and Technology (NTNU), Trondheim, Norway

[†]Department of Energy Systems, SINTEF Energy Research, Trondheim, Norway

Abstract—Collective self-consumption (CSC) is a way to increase the profitability of distributed renewable energy. In CSC schemes, a sharing key is used to virtually distribute the renewable energy production. This key impacts the cost reduction for each household, and if households are flexible, it will also impact the physical grid exchange. The aim of this article is to investigate how the new CSC scheme in Norway impacts the cost reduction for energy community members and the physical exchange with the grid, by investigating equal and yearly consumption-based sharing keys. We find that the largest cost reductions are obtained for the yearly consumption-based key, especially when households are flexible. Flexibility, however, increases the maximum import at the point of common coupling by 9.6%, due to each household individually optimising against spot price variations. Comparing the grid impact and grid related costs in the different cases, we conclude that the CSC scheme is likely to shift costs over to other end-users in the grid.

Index Terms—Collective self-consumption, Regulation, Energy community, Prosumers, Sharing keys

I. INTRODUCTION

In the past two years, Norway has seen a threefold increase in the number of electricity consumers who generate their own power, also known as prosumers. A net metering scheme has been in place since 2017 for prosumers not feeding more than 100 kW into the grid, however, prosumers could not share electricity with neighbours. Also, apartment units could not share the production from joint investments since each dwelling unit had to be measured independently. This changed in October 2023, when a new scheme for collective self-consumption (CSC) up to 1 MW (AC) installed generation capacity per property took effect [1].

CSC is connected to, but not necessarily the same as, energy communities, since the enabling of CSC might lead to customers investing in joint assets. The benefit for each member of the energy community depends on how the generation is shared, through so-called sharing keys. A CSC scheme is expected to lower grid tariff costs and tax costs for the members. However, in countries with low seasonal correlation in load and solar irradiance, CSC might not necessarily give benefits to the grid, since the peak load occurs in hours of low or no solar irradiance. It is therefore important to understand how CSC schemes with different sharing keys affect the costs for the energy community members, along with how it impacts the distribution grid.

Ref. [2] proposed an energy sharing method for CSC, where each household had photovoltaic (PV) generation. The *sharing*

key was calculated from the share of each household's imported and exported electricity compared to the total imported and exported electricity. Ref. [3] optimised the sizing of a self-consumption installation for CSC in a Spanish residential building. The authors investigated three *sharing coefficients*: flat, optimal and demand-based. Ref. [4] proposed a model based on constant *distribution coefficients* for allocating self-consumed energy, as well as a constant distribution for allocating economic rights corresponding to surplus energy. Ref. [5] evaluated static and dynamic distribution coefficients in terms of maximising participants' savings. The highest savings occurred when using distribution coefficients that considered the customer's energy consumption. Refs. [6]–[8] investigated CSC in France where *keys of repartition*, calculated based on investment, load, or fixed share, were used to distribute the total generation in the community.

Although some literature has investigated CSC and sharing keys, there is no literature on CSC in Nordic countries, and none of the papers mentioned has quantified the impact on the grid. The aim of this article is to investigate how the new CSC scheme in Norway will impact the members of the energy community and the distribution grid. The main contributions are:

- Investigation of how the CSC scheme in Norway, including two static sharing keys, affects the cost distribution in the energy community.
- Quantification of how the CSC scheme impacts the distribution grid when each member optimises costs with shiftable loads.

II. METHOD AND CASE STUDY

This section describes the cases investigated, how the revenues are distributed by different sharing keys and the optimisation model used when shiftable loads are included.

A. Case overview

We investigate four cases for a fictitious energy community consisting of residents with load profiles from a real dataset from Norway, see Tab. I. There are two versions of metering considered in this paper: with or without CSC. Without CSC, PV production has its own meter and all production is directly fed to the grid. The revenue from feed-in is then divided between the members of the energy community. With CSC, PV production is virtually subtracted from the load, hour by hour, depending on the sharing key. If any household's

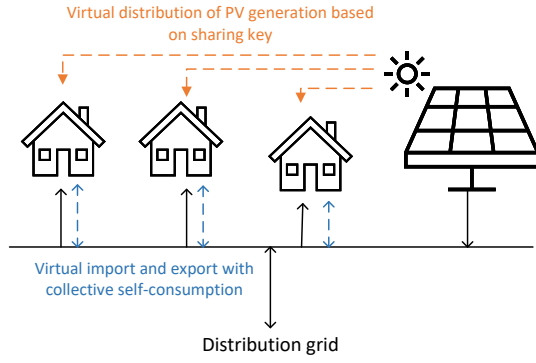


Fig. 1. Illustration of collective self-consumption from a common PV system.

net load is negative, it receives the revenue of the feed-in. Fig. 1 illustrates the difference between physical and virtual connection between the households and the PV system.

B. Method for cost calculation and grid exchange

The operational cost for each household consists of spot market cost (SC) + grid energy cost (GEC) + tax (T) + grid capacity cost (GCC) - energy revenue (ER). Most households in Norway have dynamic contracts based on the hourly spot price, as opposed to many other European countries.

1) *Cost calculation in cases without optimisation:* In cases No CSC-E, E and YC, the households are not flexible, and therefore there is no optimisation. The import and export to/from the house used to calculate the costs are virtual values, not the physical flow. The spot market cost, grid energy cost and tax for house h is calculated as:

$$SC_h + GEC_h + T_h = \sum_t (C_t^{sp} + C^{gr,e} + T) p_{t,h}^{imp} \quad (1)$$

where p_t^{imp} is the virtual import to the house in hour t , C_t^{sp} is the spot price in hour t , $C^{gr,e}$ is the energy grid tariff and T is the consumption tax and fees. The grid capacity cost is calculated as:

$$GCC_h = \sum_{m,s} C_s^{gr,p} \beta_{s,m,h}^{Step} \quad (2)$$

where $C_s^{gr,p}$ is the capacity-based grid tariff of step s and $\beta_{s,m}^{Step}$ is a binary variable which is 1 if the virtual highest monthly peak in month m is within step s . The energy revenue is calculated as:

$$ER_h = \sum_t (C_t^{sp} + R^{pr}) p_{t,h}^{exp} \quad (3)$$

where p_t^{exp} is the virtual export from the house in hour t and R^{pr} is the loss compensation from the DSO due to feed-in.

2) *Optimisation model for case YC-flex:* To find the costs, import and export in Case YC-flex, when load shifting is included, a mixed integer linear optimisation model is run for each household as given in (4)-(18). The objective function consists of the same terms as given in (1)-(3), in addition to a discomfort cost for shifting load, C^{sh} [9]. Eqs. (11)-(12) set $\beta_{s,m}^{Lim}$ equal to 1 if the maximum power of the month, p_m^{max} ,

is above or below limit L_l , by the use of a large number M . Eq. (13) determines if we are within a step s by setting $\beta_{s,m}^{Step}$ equal to 1 if two consecutive limits, $\beta_{s,m}^{Lim}$, are equal to 1. Eq. (14) ensures that only one step is 1 for each month. Eq. (5) controls the energy balance at the house's connection point to the grid, where P_t^D is the demand, P_t^{PV} is the virtual PV generation assigned to the house, $p_t^{sh,u}$ is the upward shifted load and $p_t^{sh,d}$ is the downward shifted load in hour t . Eqs. (6)-(10) limit the load shifting to a percentage, S^{max} , of the demand and keep track of the energy shifted up, $e_t^{sh,u}$, and down, $e_t^{sh,d}$. The energy shifted up and down must be equal over the course of a day, d .

$$\min \sum_t [(C_t^{sp} + C^{gr,e} + T) p_t^{imp} - (C_t^{sp} + R^{pr}) p_t^{exp} + C^{sh} p_t^{sh,d}] + \sum_{m,s} C_s^{gr,p} \beta_{s,m}^{Step} \quad (4)$$

$$p_t^{imp} - p_t^{exp} = P_t^D - P_t^{PV} + p_t^{sh,u} - p_t^{sh,d} \quad \forall t \quad (5)$$

$$p_t^{sh,u} \leq S^{max} P_t^D \quad \forall t \quad (6)$$

$$p_t^{sh,d} \leq S^{max} P_t^D \quad \forall t \quad (7)$$

$$e_t^{sh,d} = e_{t-1}^{sh,d} + p_t^{sh,d} \quad \forall t > 0 \quad (8)$$

$$e_t^{sh,u} = e_{t-1}^{sh,u} + p_t^{sh,u} \quad \forall t > 0 \quad (9)$$

$$e_d^{sh,u} = e_d^{sh,d} \quad \forall d \in D \quad (10)$$

$$p_m^{max} - L_l \leq \beta_{l,m}^{Lim} \cdot M \quad \forall m, l \in L \quad (11)$$

$$L_l - p_m^{max} \leq (1 - \beta_{l,m}^{Lim}) M \quad \forall m, l \in L \quad (12)$$

$$\beta_{s,m}^{Step} = \beta_{s,m}^{Lim} - \beta_{s+1,m}^{Lim} \quad \forall m, s \in S \quad (13)$$

$$\sum_{s \in S} \beta_{s,m}^{Step} = 1 \quad \forall m \quad (14)$$

$$\beta_{s,m}^{Step} \in \{0, 1\} \quad \forall m, s \in S \quad (15)$$

$$\beta_{l,m}^{Lim} \in \{0, 1\} \quad \forall m, l \in L \quad (16)$$

$$p_t^{imp}, p_t^{exp}, p_t^{sh,d}, p_t^{sh,u}, e_t^{sh,d}, e_t^{sh,u} \geq 0 \quad \forall t \quad (17)$$

$$p_m^{max} \geq 0 \quad \forall m \quad (18)$$

3) *PV share:* What separates cases No CSC-E, E and YC are the PV profiles assigned to each household, which again affect the virtual imported and exported power. The PV profiles assigned to each household h in hour t is calculated as $P_{h,t}^{PV} = P_t^{PV,tot} \cdot \gamma_h$, where $P_t^{PV,tot}$ is the total PV generation in hour t and γ_h is the sharing key of household h . The equal key is calculated as $\gamma^E = \frac{1}{N_h}$, where N_h is the number of households. In this case, $\gamma^E = 1/423$, since we have 423 households. The yearly consumption key for household h is calculated as $\gamma_h^{YC} = \frac{E_h^Y}{\sum_h E_h^Y}$, where E_h^Y is the annual energy consumption of household h .

4) *Fairness indicator:* The impact of the sharing keys on energy community members' economy is assessed using a fairness indicator. The EI, inspired by the Gini Index and adapted for local energy markets in [10], assesses revenue distribution equality among community members:

$$EI = 1 - \left(\frac{2 \sum_{h=1}^{N_h} h I_h}{N_h \sum_{h=1}^{N_h} I_h} - \frac{N_h + 1}{N_h} \right) \quad (19)$$

TABLE I
OVERVIEW OF CASES

Case	CSC	Sharing key	Load shifting	PV	Description
Ref.	-	-	-	-	Reference case without any new technology or sharing.
No CSC-E	-	Equal	-	✓	PV production is fed directly to the grid and the revenues are shared equally.
E	✓	Equal	-	✓	PV production is shared equally between members, and each member pays for their virtual net load.
YC	✓	Yearly cons.	-	✓	PV production is shared depending on yearly consumption, and each member pays for their virtual net load.
YC-flex	✓	Yearly cons.	✓	✓	Members can shift 20% of their load. PV production is shared depending on yearly consumption, and each member pays for their virtual net load.

The indicator is defined between 0-1. A value of 1 signifies a perfectly fair situation where everyone gets equal revenue. Conversely, a value of 0 suggests an entirely unfair situation where one member receives all the revenue. Each household's total revenue, I_h , is the cost difference between the investigated case and the reference case, representing the advantage of community participation. In (19), the revenue is sorted in ascending order. Hence, the index h is reassigned such that 1 corresponds to the member with the lowest revenue, and N_h corresponds to the member with the highest revenue.

C. Input data

The case study is a fictitious energy community of 423 households connected to an upstream power grid with no restrictions on import or export. The load profiles are hourly profiles retrieved from a Norwegian dataset from the Oslo region [11] for 2021¹. The PV profile is retrieved from Renewables Ninja [12] for location Oslo, year 2019, tilt 40°, and scaled to 1 MWp since this is the largest capacity allowed in the CSC scheme. We assume that 1/3 of the PV panels are facing south, 1/3 southeast and 1/3 southwest. Fig. 2 shows the annual load and PV generation, which amounts to 5064 MWh and 968 MWh, respectively.

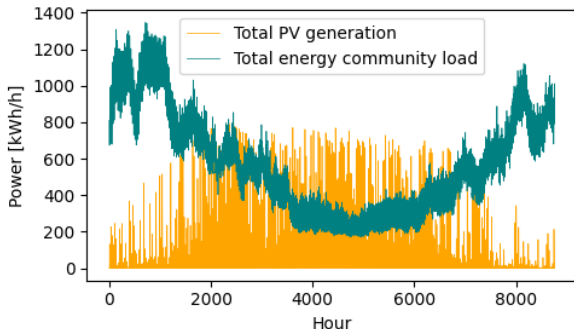


Fig. 2. Total load and PV generation in energy community

The hourly spot price is from Nordpool, NO1, Oslo region, for 2021. The grid tariff is from the DSO Elvia, and includes

¹The households consist of 58% apartments, 18% linked houses, 15% detached houses, 5% semi-detached houses and 4% other.

fees and taxes [13]². The capacity-based grid tariff depends on which step the monthly peak falls within (see Tab. II) [13]. It

TABLE II
CAPACITY-BASED GRID TARIFF

Step, s	1	2	3	4	5
Peak power limits, L_l [kWh/h]	0-2	2-5	5-10	10-15	15-20
Cost, $C_s^{gr,p}$ [€/month]	12.5	20	32.5	45	57.5

is assumed that 20% of the load can be shifted in each hour, at a discomfort cost of 0.02 €/kWh [9].

III. RESULTS AND DISCUSSION

In this section, we display one summer day for one household, to visualise the impact of the different sharing keys. Next, we summarise the annual grid exchange and the cost distribution within the energy community. Finally, we present the fairness indicators for each case.

A. Physical and virtual grid exchange

Fig. 3 shows the physical and virtual grid exchange in all cases for one summer day, for the house with the highest electricity consumption. The physical import to the house is the same in cases No CSC-E, E and YC (gray line), but changes in case YC-flex due to shiftable loads (red line), where the peak load is increased. Since the capacity-based grid tariff is calculated from which step the consumption is within, this house can profitably optimise against spot price, as long as its peak stays below 5 kWh/h. The load is significantly reduced for the virtual grid exchange in case E (blue line), and there is export in some hours. In case YC (orange line), the virtual exchange for this house shows a high export reaching 5 kW. In case YC-flex, which includes load shifting, the virtual exchange is altered slightly, compared to case YC. A house with lower electricity consumption would have much higher virtual export, and be able to shift a lower amount of load in case YC-flex.

Tab. III shows the annual physical grid exchange at the point of common coupling (PCC) of the energy community.

²Jan-Mar Weekdays 06-22: 3.529 €/kWh, Night and weekend: 2.904 €/kWh. Apr-Dec Weekdays 06-22: 4.364 €/kWh, Night and weekend: 3.739 €/kWh. Prosumers are also compensated for loss reduction: 0.6 €/kWh in Apr-Oct and 0.85 €/kWh in Nov-Mar [14].

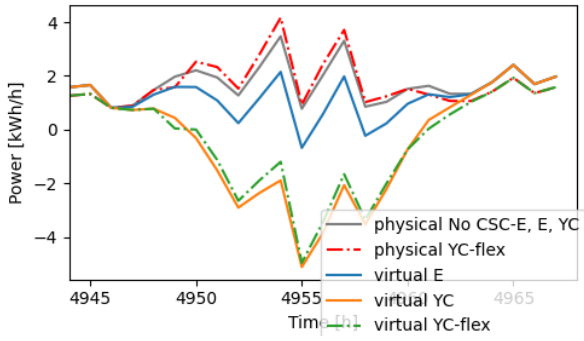


Fig. 3. Grid exchange for house with the highest consumption, 26 July

The values of cases No CSC-E, E and YC are equal, since the load and PV generation are equal, and the only change is the calculation of costs and revenues. For these cases, the maximum import remains unchanged compared to the reference case, since the PV generation is low in winter (see Fig. 2). The sum of imported electricity is reduced by 15.7% in the cases without load shifting and 16.3% in case YC-flex. The load shifting increases the self-consumption rate (SCR) to 85%, compared to 82% in the other cases. The maximum import, on the other hand, is increased to 1.48 MWh/h, a 9.6% increase compared to the reference case. This happens because the households shift their load to an hour of low spot price, leading to a higher total peak. The maximum export is 0.51 MWh/h for cases without load shifting, and decreases by 7.8% for case YC-flex. Hence, when each household uses flexibility to minimise individual costs, it leads to a higher maximum import and a lower maximum export. The increased maximum import is a result of the flexibility in combination with spot price variations and step-wise increase in capacity-based grid tariffs, thus not a direct result of PV generation or the CSC scheme.

TABLE III
SUMMARY OF ANNUAL GRID EXCHANGE AND SCR AT PCC

Case	Sum imp. [MWh]	Sum exp. [MWh]	Max. imp. [MWh/h]	Max. exp. [MWh/h]	SCR [%]
Ref.	5,065	0	1.35	-	-
No CSC-E, E, YC	4,269	173	1.35	0.51	82
YC-flex	4,240	144	1.48	0.47	85

B. Energy community operational costs

Tab. IV shows the total operational costs for the energy community, split in the different cost components. The costs in case No CSC-E are lower than in the reference case, due to the revenue from PV generation. In case E, the costs are reduced mainly due to a reduction in spot market cost, grid energy cost and taxes. Comparing the two keys (case E and YC), the yearly consumption key leads to lower spot market costs, taxes and energy revenues. Hence, sharing the PV generation equally leads to higher energy community costs than sharing it based

on yearly consumption. Case YC-flex has the lowest cost, since the shiftable loads can increase the self-consumption of PV generation for each household and optimise against spot price and grid tariffs, thereby reducing the total costs with 16% compared to the reference case, and by 9% compared to case E. Tab. IV shows that the total grid related costs of the

TABLE IV
TOTAL ANNUAL OPERATIONAL COST FOR ENERGY COMMUNITY, SPLIT INTO DIFFERENT COST COMPONENTS [k€]

Cost component	Ref.	No CSC-E	E	YC	YC-flex
SC	394	394	355	350	339
GEC	79	79	68	67	66
T	208	208	184	180	177
ER	0	-66	-23	-17	-14
GCC	115	115	113	113	100
Total	796	731	698	693	668

energy community are decreased by the introduction of a CSC scheme, relative to the case without solar (Ref.) and the case without CSC (No CSC-E). The peak export is decreased in case YC-flex, but the peak import is increased or unchanged. Therefore, it is likely that the CSC scheme, with the present grid tariffs, will lead to grid costs being shifted over to other end-users.

Fig. 4 shows a scatter plot of the absolute cost reduction for each house compared to the reference case for all cases, plotted by yearly consumption of the house. In case No CSC-E, all houses experience an equal reduction in costs, since the revenue from the PV generation is equally shared between the houses independent of the yearly consumption. Case E exhibits an even larger cost reduction for all houses, where houses with higher consumption have a slightly higher cost reduction. Case YC, on the other hand, gives a much larger spread in cost reductions, and many houses have a smaller cost reduction than in case No CSC-E. We also see that there are several large consumers which experience a significant cost reduction, up to 1000 €. Case YC-flex is similar to case YC, but here some houses experience cost reductions of up to 1200 €.

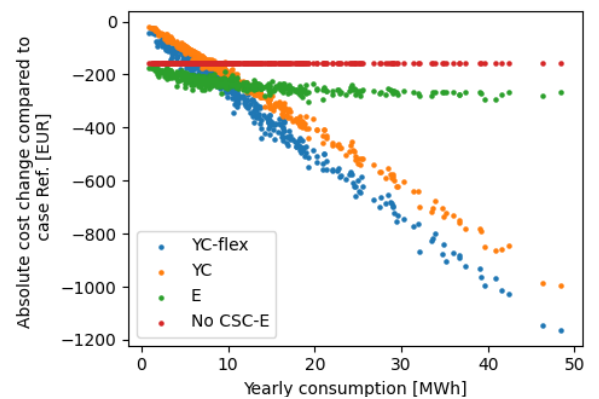


Fig. 4. Cost reduction versus yearly consumption of house.

Fig. 5 shows the relative change in total cost for each household, compared to the reference case. The relative change

presents a different picture than Fig. 4. Here, cases No CSC-E and E give the largest spread in cost reduction, where some households experience a cost reduction of up to 60%. Cases YC and YC-flex, on the other hand, have a more narrow range with all households within a 25% cost reduction.

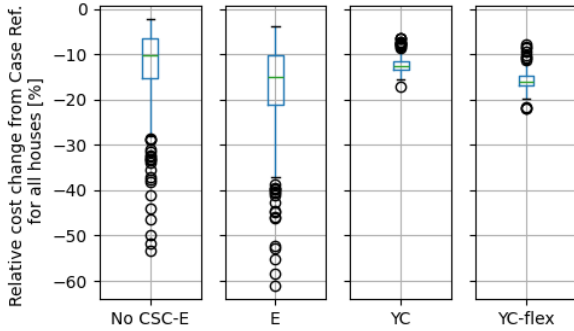


Fig. 5. Box plot of cost reduction for all houses.

C. Fairness evaluation

As Figs. 4 and 5 show, the results can support different stories depending on how fairness is defined. Here, the indicator presented in Section II-B4 is used to assess the economic fairness of the different keys. The values for all five cases can be found in Tab. V. The indicators are calculated based on the annual results. The two keys represent two different views of fairness: equality (case E) and equity (case YC). The fairness indicators, which are defined based on the same ideas, can thus be used to evaluate how well each key achieves its own definition of fairness, rather than comparing the two. The first fairness indicator, EI^{absolute} , is based on the equality of the absolute revenue among community members, corresponding to and confirming the results in Fig. 4. The second indicator, EI^{relative} , is based on the relative cost reduction among the community members, as in Fig. 5, thus measuring equity-based fairness. In terms of equality, the Equal key performs well both with and without CSC. The equity-based key does not achieve absolute fairness in any scenario, although it comes close. However, to get the full picture of the fairness, the investment costs and their distribution would need to be included. The fairness evaluation here is a good starting point for considering the distribution of investment costs and revenues when introducing shared assets within a CSC scheme. This is relevant also when introducing and comparing alternative sharing keys, as these could reflect the distribution of investment costs.

TABLE V
FAIRNESS INDICATORS FOR EACH CASE

	Ref.	No CSC-E	E	YC	YC-flex
EI^{absolute}	-	1	0.936	0.622	0.639
EI^{relative}	-	0.66	0.71	0.93	0.94

IV. CONCLUSION

The aim of this article was to investigate how the new CSC scheme in Norway will impact the members of the

energy community and the distribution grid. We quantified the individual cost savings for each household and the grid exchange at the point of common coupling for different PV generation sharing keys. The cost savings reflect how much the energy community or household could be willing to invest in shared PV generation assets. We found that the largest cost reductions were obtained for the yearly consumption-based sharing key, especially when households were assumed to be flexible. When introducing flexibility, however, the maximum import at the point of common coupling was increased by 9.6%, due to each household individually optimising against spot price. The maximum export, on the other hand, was decreased by 7.8%. Comparing the grid impact and grid related costs, we conclude that it is likely that the CSC scheme, with the present grid tariffs, will lead to grid costs being shifted over to other end-users.

ACKNOWLEDGEMENT

This work has been supported by the consortium of the research project "FINE - Flexible Integration of Local Energy Communities into the Norwegian Electricity Distribution System" (308833), and the research centre FME CINELDI (257626), both funded by the Research Council of Norway.

REFERENCES

- [1] Ministry of Energy, "Forskrift om kraftomsetning og netjenester [eng.: Regulations regarding electricity trade and grid services]," <https://lovdata.no/forskrift/1999-03-11-301/3-12>, accessed: 01-02-2024.
- [2] V. Z. Gjorgievski, S. Cundeva, N. Markovska, and G. E. Georghiou, "Virtual net-billing: A fair energy sharing method for collective self-consumption," *Energy*, vol. 254, p. 124246, Sep. 2022.
- [3] A. J. Gil Mena et al., "Analysis and optimisation of collective self-consumption in residential buildings in Spain," *Energy and Buildings*, vol. 283, p. 112812, Mar. 2023.
- [4] J. T. Villalonga Palou et al., "Sharing approaches in collective self-consumption systems: A techno-economic analysis of the Spanish regulatory framework," *Energy Strategy Reviews*, vol. 45, p. 101055, Jan. 2023.
- [5] E. Llera-Sastresa et al., "Effect of sharing schemes on the collective energy self-consumption feasibility," *Energies*, vol. 16, no. 18, p. 6564, Jan. 2023.
- [6] J. E. Contreras-Ocaña et al., "Integrated planning of a solar/storage collective," *IEEE Trans. Smart Grid*, vol. 12, no. 1, pp. 215–226, 2020.
- [7] A. D. Mustika et al., "A two-stage management strategy for the optimal operation and billing in an energy community with collective self-consumption," *Applied Energy*, vol. 310, p. 118484, 2022.
- [8] H. Camblong et al., "Photovoltaic energy sharing: Implementation and tests on a real collective self-consumption system," *Heliyon*, vol. 9, no. 12, p. e22252, Dec. 2023.
- [9] E. F. Bødal et al., "Demand flexibility modelling for long term optimal distribution grid planning," *IET Generation, Transmission & Distribution*, vol. 16, no. 24, pp. 5002–5014, 2022.
- [10] Y. Zhou, J. Wu, and C. Long, "Evaluation of peer-to-peer energy sharing mechanisms based on a multiagent simulation framework," *Applied Energy*, vol. 222, pp. 993–1022, 2018.
- [11] M. Hofmann, S. Bjarghov, and S. Nessa, "Norwegian hourly residential electricity demand data with consumer characteristics during the European energy crisis," *Data in Brief*, vol. 51, p. 109687, Dec. 2023.
- [12] S. Pfenninger and I. Staffell, "Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data," *Energy*, vol. 114, pp. 1251–1265, Nov. 2016.
- [13] Elvia, "Standard nettleietariff for forbrukerkunder [eng.: Standard grid tariffs for consumers]," <https://shorturl.at/Pa2mn>, accessed: 21-12-2023.
- [14] —, "Plusskunder – innmating av overskuddsproduksjon [eng.: Prosumers - feed-in of surplus production]," <https://shorturl.at/TXCWk>, accessed: 21-12-2023.